## 8) Equipment Inventory

SerialNo → {modelId, labId} and modelId → manufacturer

serialNO → modelId and modelId → manufacturer

SerialNo → manufacturer (via la transitivité)

serialNO → {modelId, labId, manufacturer}

- La Relation ne satisfait pas BCNF car modelId → manufacturer et modelId n'est pas un supercle

n'est pas 3NF manufacturer ∉ candidate Key = {serialNo}

Strongst Normal form : 1NF

Decomposition : modelId → manufacturer

$(modelId)^+ = \{manufacturer, modelId\}$

$R_1 = (\underbrace{modelId}_{\text{primary Key}}, manufacturer) = X^+$ BCNF

$R_2 = (modelId, \underbrace{serialNo}_{\text{primary Key}}, labId)$ BCNF

on a $F_{R_1} = \{modelId \rightarrow manufacturer\}$

$F_{R_2} = \{serialNo \rightarrow \{modelId, labId\}\}$

Donc FD are Preserved

Scanned with CamScanner

b) Project Participation

candidate Key is {pid, projectCode}

Comme

{pid, projectCode} → {pid, projectCode, role, piPid}

la Relation ne satisfait pas BCNF vu que

projectCode → piPid et projectCode is not a SuperKey

projectCode ↛ pid

N'est pas 3NF

D'où Strongest Normal form 1NF. piPid ∉ candidateKey = {pid, projectCode}

$(projectCode)^+ = \{projectCode, piPid\}$

$R_1 = (\underbrace{projectCode}_{\text{Primary Key}}, piPid)$ BCNF

$R_2 = (\underbrace{projectCode, pid}_{\text{primary Key}}, role)$ 3CNF

$F_{R_1} = \{projectCode \rightarrow piPid\}$

$F_{R_2} = \{(pid, projectCode) \rightarrow role\}$

D'où FD are Preserved

Scanned with CamScanner

C) Equipment Certification Requirment

candidate key is {modelId, certCode}

Because certCode → {certTitle, issuingAuthority}

is Not BCNF vu que CertCode n'est pas un superKéy

en effet CertCode ↛ modelId

Strongest Normal form : 1NF, 3NF

vu que CertCode ∈ Candidate key

$R_1 = (\text{CertCode})^+ = \{\text{certCode, certTitle, issuingAuthority}\}$ BCNF

$R_2 = (\text{certCode}) \cup (R - X^+) = (\text{certCode}) \cup (R - R_1) = \{\text{certCode, modelId}\}$ BCNF

$F_{R_1} = \{\text{certCode} \rightarrow \{\text{certTitle, issuingAuthority}\}\}$

$F_{R_2} = \{(\text{certCode, modelId}) \rightarrow \text{CertCode et modelId}\}$

$(F_{R_1} \cup F_{R_2})^+ = \{(\text{certCode, modelId}) \rightarrow \{\text{certTitle, issuingAuthority, certCode, modelId}\}$

, certCode → {certTitle, issuingAuthority}}

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$

D'où FD are preserved

d) Supplier Offer

candidate Key is {SuppId, consId, labId}

Because (suppId, consId, labId) → suppId → supplierName

is Not BCNF vu que suppId is not a superkey

en effet suppId ↛ consId. or labId

Strongest Normal Form : 1NF, 3NF

Comme suppId ∈ candidate Key

$R_1 = (SuppId)^+ = \{\underbrace{SuppId}_{\text{primary Key}}, SupplierName\}$ BCNF

$R_2 = \{\underbrace{SuppId, consId, labId}_{\text{primary Key}}, unitPrice\}$ BCNF

De meme $F_{R_1} = \{ suppId \rightarrow \{suppId, suplierName\}\}$

$F_{R_2} = \{\{SuppId, consId, labId\} \rightarrow unitPrice\}$

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$

D'où FD are Preserved.

Scanned with CamScanner

## e) Reservation and Project Information

candidate Key is {resId}

Because resId → projectCode → projectTitle

et encore resId → {projectCode, madeByPid, status}

is Not BCNF vu que projectCode is not a SuperKey

en effet projectCode ↛ resId

Strongest Normal Form : 1NF

is Not 3NF Comme projectCode & candidateKey

$R_1 = (\text{projectCode})^+ = \{\underbrace{\text{projectCode}}_{\text{primary key}}, \text{projectTitle}\}$ BCNF

$R_2 = \{\text{projectCode}, \underbrace{\text{resId}}_{\text{primary key}}, \text{madeByPid}, \text{status}\}$ BCNF

De meme $F_{R_1} = \{\text{projectCode} \rightarrow \text{projectTitle}\}$

$F_{R_2} = \{\text{resId} \rightarrow \{\text{resId}, \text{projectCode}, \text{madeByPid}, \text{status}\}\}$

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$

D'où FD are Preserved.

Scanned with CamScanner